

Multiple Choice (10 marks)

1. What is the percentage of angular magnification when one views an object at 33 cm through a pane of glass 5 mm thick?
 - (a) 0.7%
 - (b) 3%
 - (c) 0.1%
 - (d) 1.5%
 - (e) 1%
2. Calculate the de Broglie Wavelength of a tennis ball of mass 57 g traveling 25 m/s in meters.
 - (a) 3.0×10^{-32}
 - (b) 5.5×10^{-33}
 - (c) 1.2×10^{-30}
 - (d) 8.9×10^{-36}
 - (e) 2.3×10^{-30}
3. A 3200-lb car traveling with a speed of 60 mi/hr rounds a curve whose radius is 484 ft. Find the necessary centripetal force.
 - (a) 1200 lb
 - (b) 1800 lb
 - (c) 2000 lb
 - (d) 1000 lb
 - (e) 2400 lb
4. A barometer that uses water instead of mercury will be
 - (a) more fragile
 - (b) shorter
 - (c) heavier
 - (d) equal in height
 - (e) taller
5. Kirkwood gaps are observed in the main asteroid belt including at the position(s) where:

- (a) asteroids would orbit with a period that is three times that of Jupiter
- (b) asteroids would orbit with a period three times that of Mars
- (c) asteroids would orbit with a period twice that of Mars
- (d) asteroids would orbit with a period twice that of Jupiter
- (e) A and B

True / False (10 marks)

Answer True or False

6. True or False: The displacement of hiker 2 is 18 km, which is half the distance covered by hiker 1 multiplied by 6.
7. True or False: One astronomical unit (AU) is commonly approximated to be around 170 million kilometers, which is often used in astronomical calculations.
8. True or False: Knowing both the actual brightness and apparent brightness of an object allows you to estimate its distance from your location.
9. True or False: Orange light bends more than green light when white light passes through a prism, leading to a greater dispersion.
10. True or False: An object in free fall will cover a distance of h during the second second of its descent if released from rest.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A proton falls toward the Earth under gravitational influence, converting its kinetic energy into a quantum of light upon impact. Calculate the frequency of this light and discuss the underlying principles involved in this energy transformation.
12. Discuss the principles behind calculating the frequency of the hydrogen transition from the $n = 101$ to $n = 100$ energy level. Include relevant equations and analyze the implications of your findings.

End of Paper